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## RESEARCH ARTICLE

# High-Performance Lung Disease Identification and Explanation Using a ReciproCAM-Enhanced Lightweight Convolutional Neural Network

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**ABSTRACT** Lung diseases significantly contribute to global morbidity and mortality, underscoring the importance of accurate and efficient diagnostic tools. This study introduces a lightweight two-dimensional convolutional neural network (LW-2DCNN) integrated with ReciproCAM, an explainable artificial intelligence (XAI) technique, specifically designed to classify multiple pulmonary conditions from chest computed tomography (CT) scans. Our model effectively addresses limitations of existing diagnostic methods by enhancing computational efficiency, diagnostic accuracy, and interpretability. Utilizing a curated dataset comprising 10,000 CT images from a Korean population, the LW-2DCNN achieved exceptional accuracy of 99.80% across five diagnostic categories: normal, lung cancer, pneumothorax, pneumonia, and interstitial lung disease (ILD). Comparative analyses demonstrated superior performance relative to established benchmark models, including EfficientNetB0, ResNet50, InceptionV3, DenseNet121, NAS-NetMobile, MobileNetV2, and Xception. The integrated ReciproCAM technique significantly enhanced clinical interpretability by highlighting the precise anatomical regions influencing the model's diagnostic decisions. Ablation studies validated the optimal choice of Adam optimizer and demonstrated the model's computational feasibility, predicting and explaining 1000 images in approximately 150 seconds on a CPU-only device. This indicates strong potential for deployment in resource-constrained clinical settings.

**INDEX TERMS** Computer aid system, explainable AI, lightweight model, lung diseases.

## I. INTRODUCTION

Lung diseases impose considerable challenges on global healthcare systems, significantly affecting public health through high morbidity and mortality rates. Based on data reported by the WHO, approximately 5% of global mortality in 2021 was attributed to chronic obstructive pulmonary disease (COPD), making it the fourth most common cause of death worldwide [1]. Moreover, lung cancer continues to be the deadliest cancer type globally, leading to roughly 1.8 million fatalities in 2020 [2]. Given these statistics, the

precise diagnosis of pulmonary conditions is essential for guiding treatment decisions and assessing the necessity of hospitalization.

The diagnosis and classification of lung diseases traditionally depend on medical imaging techniques such as chest X-rays (CXRs) and computed tomography (CT) scans [3]. However, even highly trained radiologists may encounter challenges in accurately interpreting these images. Unlike CXRs, chest CT scans provide detailed images in cross-sectional slices, offering three-dimensional insights into anatomical details. Due to its enhanced sensitivity and reduced variability between observers when identifying pulmonary abnormalities, this imaging method has become the

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method of choice for both diagnosis and surveillance of diverse lung diseases. Moreover, the improved availability of CT machines, reduced costs, lower radiation doses, and continual technological enhancements have led to a marked increase in CT examinations performed each year. However, the increasing workload, coupled with a shortage of radiologists and the risk of diagnostic fatigue, underscores the need for alternative interpretative solutions. Hence, implementing a robust computer-aided detection (CAD) system for chest CT analysis could streamline radiologists' workflow, minimize interpretation time, and improve diagnostic accuracy.

While CAD systems provide a promising approach, many existing models suffer from high false-positive rates, limited accuracy, and poor generalizability. Additionally, these systems often require substantial computational resources due to their large number of parameters. Given the constraints of healthcare environments, the development of lightweight deep learning architectures is essential to ensure efficient deployment. Several studies have explored such designs; for instance, lightweight convolutional neural networks (CNN) have been successfully applied for classifying 12 echocardiographic views [4] and detecting abnormal prostate tissues [5]. MobileNets demonstrated the effectiveness of compact deep learning models for real-time mobile and embedded vision applications [6]. Similarly, EfficientNet introduced an optimized model architecture, which has been adapted for COVID-19 screening using CXRs images, highlighting its potential for medical imaging applications [7].

Despite the remarkable progress in deep learning applications for medical imaging, these models face a significant limitation due to their inherent 'black box' nature. Their intricate architectures make it difficult to interpret the rationale behind their predictions, posing a substantial challenge to their adoption and integration into clinical practice [8]. Although deep learning techniques have achieved good accuracy in identifying lung diseases from medical images, their lack of transparency remains a critical obstacle. For clinicians to confidently use AI-driven diagnostic tools, it is essential to ensure the interpretability and reliability of the model's decision-making processes.

To develop a dependable CAD solution for chest CT analysis, this research introduces an explainable AI (XAI) framework capable of classifying multiple lung diseases including lung cancer, pneumothorax, pneumonia, and interstitial lung disease (ILD). At its core is a lightweight 2D convolutional neural network (LW-2DCNN) coupled with ReciproCAM [9], a lightweight gradient-free class activation map technique designed to generate interpretable saliency maps. By highlighting critical regions within CT scans, ReciproCAM clarifies the factors underpinning the model's diagnoses, thus fostering transparency and clinician confidence. Our proposed framework not only aims to alleviate radiologists' workload and diagnostic fatigue but is also structured for use in healthcare environments with constrained computational resources.

The key contributions of our study are as follows:

- A customized LW-2DCNN model was developed for identifying multiple lung diseases using a Korean population chest CT dataset.
- A thorough evaluation was conducted to determine the LW-2DCNN architecture's performance, specifically focusing on accuracy and computational efficiency, through comparative analyses against 7 transfer learning frameworks.
- ReciproCAM was incorporated into the LW-2DCNN framework to illuminate the reasoning behind diagnostic outcomes, thereby promoting transparency and clinical applicability.
- The execution time of the proposed XAI framework was evaluated on both GPU-enabled and CPU-only computing environments to demonstrate its performance in settings with limited computational resources.

## II. LITERATURE REVIEW

Deep learning has emerged as a powerful means of interpreting and classifying medical images, particularly in the context of lung disease detection and classification [10], [11], [12]. Although numerous architectures and classifiers have been proposed, developing a model that pairs high diagnostic accuracy with low computational requirements continues to pose a substantial challenge. This section presents an extensive discussion of various methodologies employed for lung disease classification, examining their performance metrics, underlying parameters, advantages, and limitations.

Deep learning methods have garnered significant attention for categorizing pulmonary conditions. For example, Masood et al. [13] achieved 93.3% diagnostic accuracy in identifying lung cancer using a CNN model trained on CT images. Similarly, Anthimopoulos et al. [14] reported an 80% accuracy rate for classifying interstitial lung diseases. Pathak et al. [15] utilized a deep neural network based on bidirectional LSTM architectures for identifying COVID-19 using chest CT imagery, while Liu et al. [16] developed a Structural Attention Graph Neural Network for diagnosing COVID-19, integrating multiple data streams such as radiomic features derived from CT scans, implicit pulmonary structural characteristics, and clinical patient records. This method not only assesses COVID-19 severity but also predicts the timeline for progression from mild to severe cases. Despite these successes, deep learning systems often require substantial computational resources and lengthy training periods.

Efforts to enhance efficiency have led to the emergence of lightweight models in healthcare applications. Ukwandu et al. [17] fine-tuned MobileNetV2 to identify COVID-19 and viral pneumonia in CXRs. Trivedi and Gupta [18] also modified MobileNet to detect pneumonia, showing high accuracy, while Mothkur and Veerappa [19] adopted deep networks for lung cancer recognition in CT data. Huang and Liao [20] introduced LightEfficientNetV2 for both CXRs and CT datasets. Additionally,

Hussein et al. [21] proposed two lightweight CNN architectures for binary (COVID-19 vs. Normal) and multiclass (COVID-19, viral pneumonia, or Normal) classification, minimizing the number of trainable parameters by using ELU activation. For smaller datasets, Abdullah et al. [22] employed Bayesian deep learning methods to detect breast cancer, offering an alternative to CNN-based approaches. However, lightweight models may face generalization challenges, necessitating a careful balance between model size, accuracy, flexibility, and interpretability to optimize performance while mitigating overfitting.

Another influential methodology for lung disease recognition is transfer learning, which utilizes pre-trained models to improve classification outcomes. Agarwal and Arya [23] combined deep features from ResNet50, InceptionV3, and Xception to diagnosis pneumonia and COVID-19 in CXRs, attaining 94.05% accuracy. Along similar lines, Chang et al. [24] leveraged transfer learning to diagnose tuberculosis cultures and reached 99% precision and 98% recall for non-negative samples, even with limited data. While these approaches demonstrate substantial benefits, they may require high computational resources, which can sometimes make training models from scratch a more practical option.

Despite their impressive accuracy, deep learning models frequently resemble “black boxes,” concealing the basis for their decisions. This opacity has prompted growing interest in XAI strategies designed to promote transparency. Class Activation Mapping (CAM) [25] and Grad-CAM [26] offer visual representations of model reasoning, although they can be hindered by architecture-specific constraints and reliance on gradient computation. Score-CAM [27] addressed the latter shortcoming by removing gradient dependence, but at the cost of increased processing times. ReciproCAM [9], introduced to overcome these performance bottlenecks, uses spatial masking of feature maps and leverages interactions between activation patterns and model outputs to generate saliency maps. It achieves a processing speed near that of Grad-CAM while operating roughly 148 times faster than Score-CAM. However, previous studies on lightweight lung disease classification models [17], [18], [19], [20], [21], [22] have generally not incorporated XAI techniques to enhance interpretability. Some recent efforts [28], [29] have integrated Grad-CAM into lightweight models, but they were limited to a single lung disease, such as COVID-19 or lung cancer, and did not account for the computational efficiency of XAI techniques alongside lightweight classification models.

As outlined above, while various approaches have shown promise in detecting pulmonary diseases from medical images, there remains a pressing need for further research to enhance their performance and diagnosis ability on multiple lung diseases. Moreover, the computational intensity of deep learning models poses a significant barrier to their practical implementation in clinical environments. Additionally, there is still a scarcity of studies developing XAI-integrated lightweight models capable of classifying multiple pulmonary conditions. To address this existing

research gap, this study introduces an interpretable artificial intelligence framework that combines a computationally efficient LW-2DCNN architecture and ReciproCAM visualization technique, designed specifically for diagnosing various pulmonary diseases such as lung cancer, pneumothorax, pneumonia, and ILD. The proposed framework aims to introduce a reliable AI-driven diagnostic tool into resource-constrained healthcare settings, thereby reducing radiologists' workload and mitigating diagnostic fatigue.

### III. MATERIALS AND METHODOLOGY

#### A. DATASET

In this research, we utilized an open-access dataset consisting of 10,000 chest CT images obtained from AIHub [30]. The dataset comprises images collected from multiple institutions, including the Yonsei University Office of Research Affairs and University-Industry Foundation, Kosin University Gospel Hospital, and the National Cancer Center, with data synthesis conducted by Promedius Incorporated. To safeguard patient privacy, all images were preprocessed to remove personal information before being publicly released on AIHub. The dataset was developed to improve access to high-quality medical imaging data in South Korea while maintaining stringent privacy standards.

The chest CT images in this dataset were of high resolution ( $512 \times 512$  pixels) and were categorized into five groups: normal (5000 images), lung cancer (1250 images), pneumothorax (1250 images), pneumonia (1250 images), and interstitial lung disease (ILD) (1250 images). Fig. 1 presents representative examples of images from each lung disease category included in the dataset.

#### B. PREPROCESSING

For the purposes of model development and evaluation, we divided our collected data into three subsets, allocating 80% for training, 10% for validation, and 10% for testing. The training set consisted of 8000 images, distributed as follows: normal (4000 images), lung cancer (1000 images), pneumothorax (1000 images), pneumonia (1000 images), and interstitial lung disease (ILD) (1000 images). The validation and test sets each contained 1000 images, with 500 normal cases and 125 images per disease category (lung cancer, pneumothorax, pneumonia, and ILD). For the model training process, the categories were label encoded as follows: normal (class 0), lung cancer (class 1), pneumothorax (class 2), pneumonia (class 3), and ILD (class 4).

To ensure the model focused exclusively on the lung region while eliminating irrelevant background information, a contour detection technique was employed to isolate and crop the lung area. This preprocessing workflow involved the following steps:

- **Grayscale Conversion and Smoothing:** Each image was converted to grayscale to simplify processing. A Gaussian blur with a  $5 \times 5$  kernel was implemented to diminish image noise and facilitate accurate contour extraction.

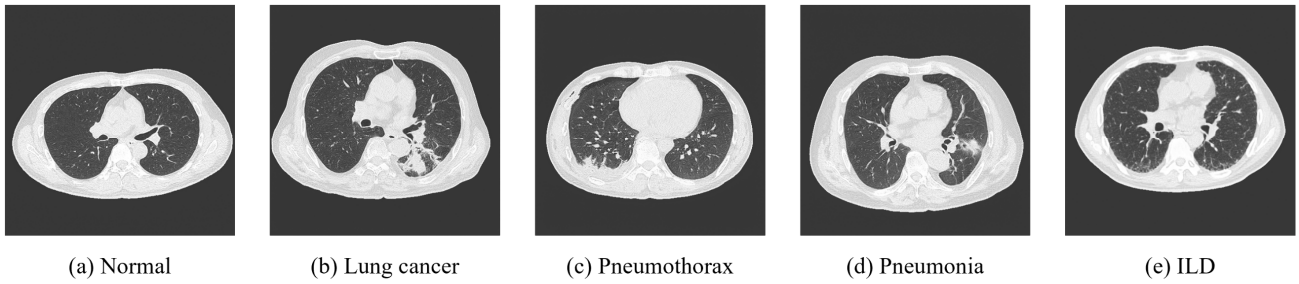


FIGURE 1. Example images from each lung disease category included in the dataset.

- **Thresholding:** Otsu’s automatic thresholding technique was employed to convert grayscale images into binary form by dynamically selecting the optimal threshold value, effectively improving the distinction between lung regions and background areas.
- **Morphological Operations:** The binary image underwent erosion (two iterations) to eliminate small noise artifacts, followed by dilation (two iterations) to restore essential structures. This process improved the accuracy and robustness of contour extraction.
- **Contour Extraction:** Contours were detected in the binary image, with the largest contour which was presumed to correspond to the lung region, was selected based on its area.
- **Cropping Using Extreme Points:** The extreme points of the largest contour (leftmost, rightmost, topmost, and bottommost) were identified, enabling precise cropping of the original image to retain only the lung region. The cropped image was then resized to a fixed dimension of  $224 \times 224$  pixels using cubic interpolation to standardize input sizes for model training.

Fig. 2 provides an example of an image following the cropping step.



FIGURE 2. An image after preprocessing.

### C. PROPOSED LW-2DCNN ARCHITECTURE

In this study, we develop an efficient LW-2DCNN architecture tailored explicitly for the classification of pulmonary conditions using chest CT imagery. The proposed LW-2DCNN design emphasizes optimized computational

TABLE 1. The proposed 2D-CNN architecture.

| Layer                  | Output shape         | Param # |
|------------------------|----------------------|---------|
| InputLayer             | (None, 224, 224, 3)  | 0       |
| Conv2D (1)             | (None, 224, 224, 32) | 8       |
| MaxPooling2D (1)       | (None, 112, 112, 32) | 0       |
| BatchNormalization (1) | (None, 112, 112, 32) | 128     |
| Dropout (1)            | (None, 112, 112, 32) | 0       |
| Conv2D (2)             | (None, 112, 112, 64) | 18496   |
| MaxPooling2D (2)       | (None, 56, 56, 64)   | 0       |
| BatchNormalization (2) | (None, 56, 56, 64)   | 256     |
| Dropout (2)            | (None, 56, 56, 64)   | 0       |
| Conv2D (3)             | (None, 56, 56, 128)  | 73856   |
| MaxPooling2D (3)       | (None, 28, 28, 128)  | 0       |
| BatchNormalization (3) | (None, 28, 28, 128)  | 512     |
| Dropout (3)            | (None, 28, 28, 128)  | 0       |
| Conv2D (4)             | (None, 28, 28, 256)  | 295168  |
| MaxPooling2D (4)       | (None, 14, 14, 256)  | 0       |
| BatchNormalization (4) | (None, 14, 14, 256)  | 1024    |
| Dropout (4)            | (None, 14, 14, 256)  | 0       |
| Conv2D (5)             | (None, 14, 14, 512)  | 1180160 |
| MaxPooling2D (5)       | (None, 7, 7, 512)    | 0       |
| BatchNormalization (5) | (None, 7, 7, 512)    | 2048    |
| Dropout (5)            | (None, 7, 7, 512)    | 0       |
| Flatten (5)            | (None, 25088)        | 0       |
| Dense (1)              | (None, 256)          | 6422784 |
| BatchNormalization (6) | (None, 256)          | 1024    |
| Dropout (6)            | (None, 256)          | 0       |
| Dense (2)              | (None, 128)          | 32896   |
| BatchNormalization (7) | (None, 128)          | 512     |
| Dropout (7)            | (None, 128)          | 0       |
| Output (Dense)         | (None, 5)            | 645     |

efficiency and scalability, adeptly capturing nuanced pathological features critical to diagnostic accuracy. A summary of the learning parameters for the proposed LW-2DCNN architecture is provided in Table 1.

The network begins with an input layer accepting standardized chest CT images of dimensions  $224 \times 224 \times 3$ . These RGB channels are derived from replicated grayscale images to ensure consistent preprocessing and computational handling.

Subsequently, the architecture comprises five sequential convolutional blocks. Each block consists of convolutional, max-pooling, batch normalization, and dropout layers arranged systematically to sequentially capture and refine increasingly abstract and informative characteristics from input data. The first convolutional block employs 32 filters with dimensions  $3 \times 3$ , followed by a max-pooling layer ( $2 \times 2$  pool size) to reduce spatial dimensions, batch normalization for stabilizing training, and dropout at a rate of 0.1 to prevent overfitting. The second block follows an analogous structure but increases filter count to 64, preserving the arrangement of pooling, batch normalization, and dropout. The third convolutional block escalates to 128 filters, continuing the established pattern of pooling, normalization, and dropout. In the fourth and fifth blocks, the filter counts expand further to 256 and 512, respectively, consistently maintaining the standardized combination of convolution, pooling, batch normalization, and dropout procedures.

After completing the feature extraction process, the resulting multidimensional feature maps from the last convolutional stage are reshaped into a one-dimensional representation to serve as input for fully connected layers. The initial dense layer contains 256 units activated using the ReLU function, followed by batch normalization and dropout at a rate of 0.3. Subsequently, another dense layer featuring 128 units is applied, similarly utilizing ReLU activation, batch normalization, and dropout regularization at a rate of 0.3.

The LW-2DCNN model concludes with an output layer containing five neurons corresponding to distinct pulmonary disease categories. The final layer employs softmax activation to produce probability-based predictions among several classification categories.

The compilation process utilized the Adam optimization algorithm due to its dynamic adjustment capabilities and efficiency in optimizing learning parameters. For loss evaluation, sparse categorical cross-entropy was selected, being well-suited for multi-category prediction tasks, while classification accuracy served as the principal criterion for performance assessment. To enhance training efficacy, several advanced optimization techniques were applied. Model checkpointing was implemented using the ModelCheckpoint callback, which saves the best model weight whenever validation loss improves, thus preserving the highest-performing model for evaluation and deployment. To optimize convergence, the ReduceLROnPlateau callback was employed to track the validation loss, reducing the learning rate by 50% whenever validation loss did not improve after three successive epochs. Furthermore, the EarlyStopping callback was implemented, terminating training if no enhancement in validation loss was observed across 12 consecutive epochs, subsequently restoring the optimal model weights. The model underwent training for 100 iterations, employing batches consisting of 64 samples each, ensuring strong generalization capabilities and employing suitable regularization methods to prevent overfitting.

#### D. BENCHMARKS

Transfer learning, a strategy that leverages pretrained models to specialize them for particular applications, has gained considerable attention in medical imaging research. To robustly evaluate our proposed LW-2DCNN model, we performed a comprehensive comparative study against seven prominent transfer learning architectures: EfficientNetB0, InceptionV3, DenseNet121, NASNetMobile, ResNet50, MobileNetV2, and Xception. These models have been previously validated and effectively applied in numerous lung disease prediction studies [31], [32], [33], [34], [35], [36], [37]. A concise overview of these benchmark models is provided below.

- EfficientNetB0 [38] was developed to optimize CNN performance through structured adjustments of the model's complexity, including modifications to its depth, breadth, and input image dimensions.
- InceptionV3 [39] is a CNN comprising 42 layers, designed around parallel convolutional and pooling operations whose outputs are concatenated to yield robust representations.
- DenseNet121 [40] incorporates 120 convolutional layers, supported by four average pooling layers and a concluding fully connected layer, emphasizing densely connected pathways to promote efficient feature propagation and mitigate gradient degradation.
- NASNetMobile [41] was developed as a computationally efficient variant of NASNet. It employs a recurrent neural network controller guided by reinforcement learning to define optimal normal and reduction cells, thus ensuring a robust yet efficient network structure.
- ResNet50 [42], a residual architecture consisting of 50 layers, employs shortcut connections to address gradient degradation issues, ensuring essential feature preservation. Its structure comprises 48 convolutional stages complemented by one max-pooling and one average-pooling operation.
- MobileNetV2 [43], a compact CNN architecture of 53 layers, includes two distinct block types: residual blocks with stride one and reduction blocks with stride two for efficient spatial dimensionality reduction. MobileNetV2 is pretrained on a diverse dataset including categories such as animals, objects, and everyday items, and is optimized specifically for  $224 \times 224$  image inputs.
- The Xception [44] model, comprising 36 convolutional layers, extends the concept behind Inception by utilizing depthwise separable convolutions, significantly reducing computational complexity while maintaining or improving accuracy.

In our experiments, we initialized each of these benchmark models with pretrained ImageNet weight. Our proposed LW-2DCNN architecture includes 28 computational layers (covering Conv2D, Dense, BatchNormalization, Dropout, Pooling, and Flatten). Hence, for each benchmark model, the top 28 computational layers were unfrozen to allow

fine-tuning, while the remaining layers stayed frozen to preserve learned representations. A customized classification head was appended to each base network, comprising a Global Average Pooling layer, a 0.3 Dropout layer to reduce overfitting, and a final five-unit dense layer using softmax for generating class probabilities. All models were then compiled with the same configurations as the LW-2DCNN, employing the Adam optimizer and sparse categorical crossentropy loss. They were trained for default 100 epochs with a batch size of 64, supplemented by ModelCheckpoint, ReduceLROnPlateau, and EarlyStopping to optimize performance. This setup ensures a fair comparison among the benchmark models and our proposed approach.

### E. RECIPROCAM BASED EXPLANATION

To improve the interpretability of the LW-2DCNN model in clinical contexts, we incorporated the ReciproCAM visualization technique. The complete workflow for ReciproCAM integration with the LW-2DCNN architecture is illustrated in Fig. 3. Specifically, ReciproCAM applies spatial masking to the target convolutional layer's feature maps, where each mask isolates an individual spatial location within the feature representation. This step results in a collection of  $H \times W$  spatially masked feature maps, each subsequently processed through the network's remaining layers. The predicted classification scores obtained from these masked maps are compiled to form a comprehensive saliency visualization. Here,  $K$  symbolizes the number of feature channels, while  $H$  and  $W$  represent the spatial dimensions - height and width, respectively.

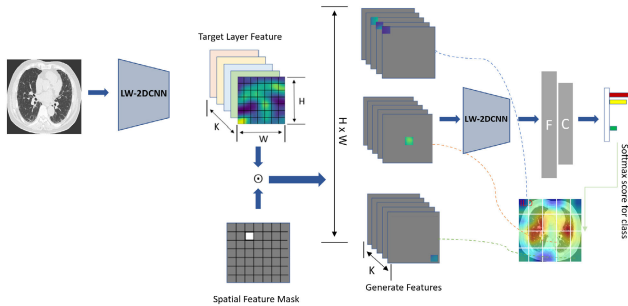


FIGURE 3. The LW-2DCNN integrated ReciproCAM's workflow.

The ReciproCAM methodology generates saliency maps by partitioning the LW-2DCNN into two primary components at a selected layer. The initial segment, denoted as  $f$ , acts exclusively as a feature extraction unit, whereas the secondary component, labeled  $g$ , encompasses the remaining network operations that yield predictions. The class-specific saliency map  $S^c$  for class  $c$  is thus mathematically defined as in (1):

$$S^c_{ReciproCAM} = \text{reshape} \left( \frac{y^c - \min(y^c)}{\max(y^c) - \min(y^c)}, (H, W) \right) \quad (1)$$

where  $y^c$  denotes the confidence scores vector of size  $N \times I$  for class  $c$ . Each entry in this vector,  $y^n_c$ , is computed as in (2):

$$y^n_c = \text{softmax}(g(f(I) \odot M^n))_c \quad (2)$$

Here,  $I$  represents the input image,  $M^n$  indicates the spatial mask applied at position  $n$ , and  $\odot$  signifies element-wise multiplication.

### F. PERFORMANCE ASSESSMENT METRICS

To rigorously assess the effectiveness of the developed classification approach, several performance indicators were calculated, namely, accuracy (Acc), recall (Rec), precision (Pre), and the F1-score (F1). Additionally, to validate the reliability and generalizability of the framework, the Matthews Correlation Coefficient (MCC) was employed as an additional statistical measure. Each metric was computed based on components extracted from a confusion matrix, specifically the true negatives (TN), true positives (TP), false negatives (FN), and false positives (FP). The detailed computational expressions for these metrics are outlined in equations (3) through (7).

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Rec} = \frac{TP}{TN + FP} \quad (4)$$

$$\text{Prec} = \frac{TP}{TP + FP} \quad (5)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (7)$$

## IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

### A. EXPERIMENTAL SETUP

All experimental procedures in this research were executed on a workstation operating under Ubuntu 22.04. Specifically, the proposed LW-2DCNN architecture, along with the benchmark architectures, was trained and tested on an advanced computing system equipped with an AMD Ryzen 9 7950X processor, an NVIDIA RTX 4090 graphics card, and 64GB of DDR5 memory. To maintain methodological consistency, all deep learning models and ReciproCAM visual explanation were implemented using TensorFlow version 2.15. Furthermore, to investigate the computational efficiency and runtime of our model on CPU-exclusive systems, we performed an ablation study on an alternative setup equipped with an AMD Ryzen 5 7600 CPU and 32GB DDR5 RAM. This secondary environment also operated under Ubuntu 22.04 and utilized TensorFlow 2.15, thereby ensuring uniformity and enabling fair comparative analyses across GPU-accelerated and CPU-based implementations.

### B. TRAINING PROCESS EVALUATION

Figs. 4 to 11 illustrate comparative analyses of the training accuracy and loss curves for the proposed LW-2DCNN architecture alongside various benchmark models. The LW-2DCNN demonstrated robust performance, with validation loss decreasing to nearly zero and validation accuracy nearing 100% within 26 epochs. Although the initial configuration permitted training up to 100 epochs, the incorporation of the EarlyStopping callback effectively curtailed unnecessary computation by halting training early at epoch 26, preserving computational efficiency and resources. Additionally, employing the ReduceLROnPlateau callback resulted in improved stability of the training process, with noticeable performance enhancements occurring after the sixth epoch, as clearly depicted in Fig. 4. This outcome underscores the effectiveness of our model fine-tuning strategy and validates the appropriateness of our architectural design.

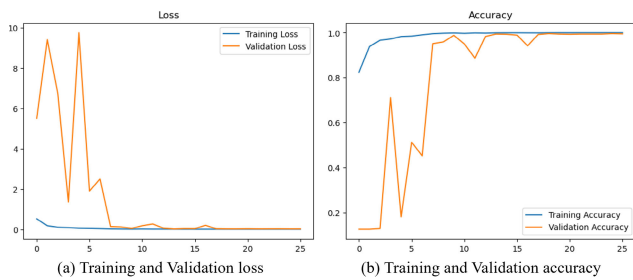


FIGURE 4. Training process of the LW-2DCNN model.

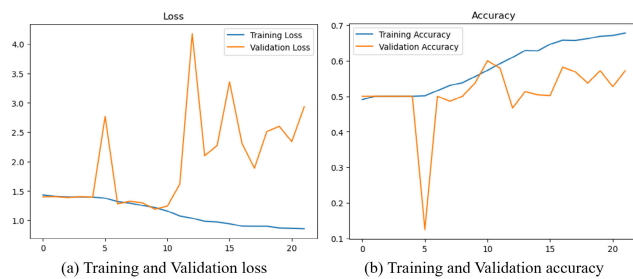


FIGURE 5. Training process of the EfficientNetB0.

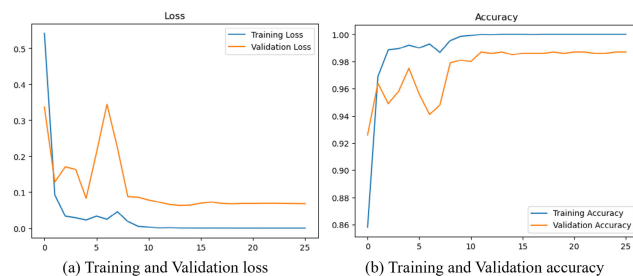


FIGURE 6. Training process of the InceptionV3.

Most benchmark models, aside from EfficientNetB0, also performed well, with validation loss below 0.25 and validation accuracy exceeding 98%. Notably, only NASNetMobile

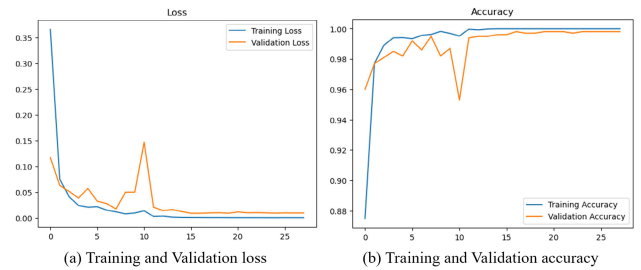


FIGURE 7. Training process of the DenseNet121.

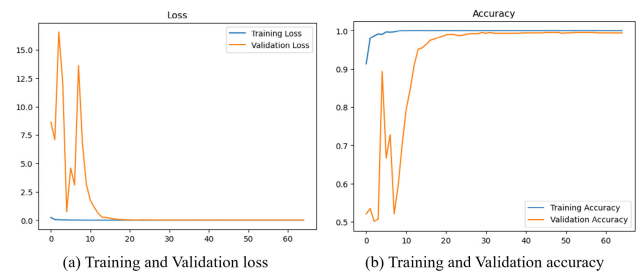


FIGURE 8. Training process of the NasNetMobile.

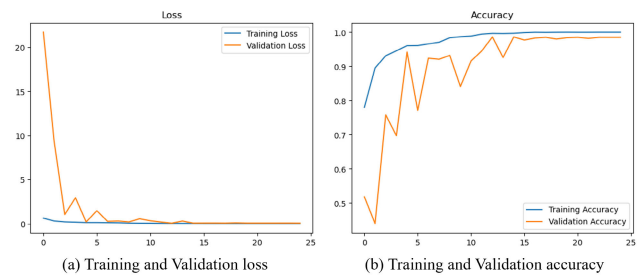


FIGURE 9. Training process of the ResNet50.

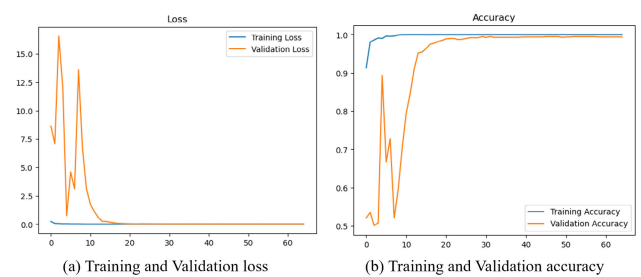


FIGURE 10. Training process of the MobileNet.

and MobileNetV2 continued training until the 100th and 60th epochs, respectively. Other benchmark architectures achieved optimal performance more rapidly, activating the EarlyStopping callback before reaching the 30th epoch due to a lack of improvement over 12 consecutive epochs. Despite these varied stopping points, the majority of models still attained high validation performance, highlighting the effectiveness of the applied training and regularization strategies across diverse deep learning architectures.

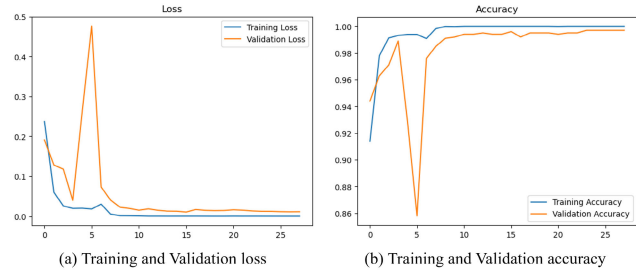


FIGURE 11. Training process of the Xception.

Fig. 12 provides a comparative analysis of the training durations across all evaluated models. According to these observations, EfficientNetB0 required the shortest training duration, approximately 1.95 minutes; however, it also exhibited the lowest validation performance. Among the models that achieved strong validation accuracy and loss metrics, the proposed LW-2DCNN model demonstrated notable computational efficiency, requiring only 2.62 minutes for training completion. This result placed it ahead of InceptionV3 and DenseNet121, which completed their training processes in 2.80 minutes and 2.86 minutes, respectively. Other benchmark architectures necessitated training times exceeding 3 minutes, with NASNetMobile being the most computationally demanding at 8.64 minutes. These findings reinforce the efficiency of the LW-2DCNN architecture, highlighting its lightweight design advantage relative to other high-performing models in the multi-class classification of pulmonary diseases from chest CT scans in this study.

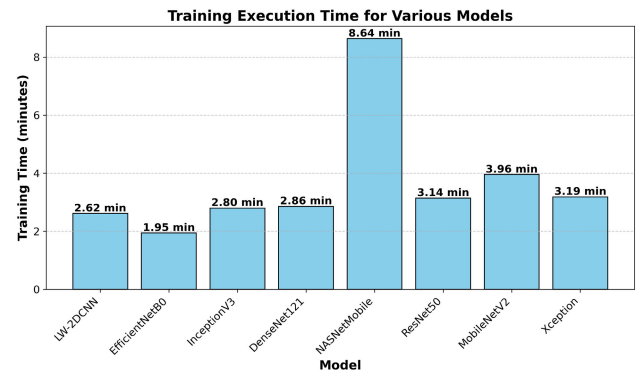


FIGURE 12. Training time comparison.

### C. PERFORMANCE ANALYSIS ON THE TEST SET

Table 2 summarizes the comparative performance metrics obtained from evaluating all trained models on the independent test dataset (unseen data). In terms of overall Acc, the proposed LW-2DCNN architecture delivered the highest result at 99.80%, surpassing all benchmark models. Closely following LW-2DCNN, the Xception and MobileNetV2 architectures each achieved Acc of 99.40%, with DenseNet121 slightly trailing at 99.00%. Other

benchmark models including ResNet50, InceptionV3, and NASNetMobile reported Acc of 98.80%, 98.50%, and 96.20%, respectively. Notably, EfficientNetB0, which demonstrated the weakest training-phase performance in validation Acc and loss, consequently yielded the lowest test-set Acc of 53.55%. These consistent results between the validation and test phases confirm that overfitting was effectively mitigated across all evaluated architectures.

TABLE 2. Performance results of all models on test set.

| Model          | Acc (%) | Rec (%) | Pre (%) | F1 (%) | MCC    |
|----------------|---------|---------|---------|--------|--------|
| LW-2DCNN       | 99.80   | 99.68   | 99.69   | 99.68  | 0.9971 |
| EfficientNetB0 | 53.55   | 25.60   | 18.29   | 20.56  | 0.2057 |
| InceptionV3    | 98.50   | 97.84   | 97.84   | 97.84  | 0.9782 |
| DenseNet121    | 99.00   | 98.40   | 98.42   | 98.40  | 0.9855 |
| NASNetMobile   | 96.20   | 94.04   | 94.44   | 94.21  | 0.9447 |
| ResNet50       | 98.80   | 98.20   | 98.20   | 98.20  | 0.9825 |
| MobileNetV2    | 99.40   | 99.04   | 99.18   | 99.09  | 0.9913 |
| Xception       | 99.40   | 99.04   | 99.05   | 99.04  | 0.9913 |

Regarding additional evaluation metrics, LW-2DCNN similarly outperformed benchmarks, achieving Rec of 99.68%, Prec of 99.69%, an F1 of 99.68%, and MCC of 0.9971. The high MCC value specifically underscores the balanced and precise predictive capability of the proposed model, irrespective of the inherent class distribution disparities within the dataset. Fig. 13 illustrates the confusion matrix for the LW-2DCNN, providing detailed insight into its predictive accuracy on unseen test cases. As depicted, the LW-2DCNN model incorrectly classified only 2 instances within the lung cancer category, correctly identifying 998 out of the 1000 test images.

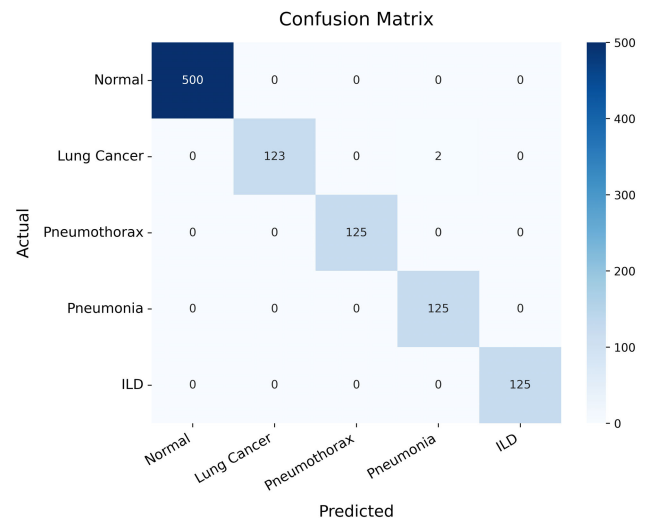


FIGURE 13. Performance of the LW-2DCNN on the test set.

Although the dataset exhibited a slight imbalance, where normal lung images (5000) outnumbered disease images (1250 per class) by a factor of 4, its high-quality curation and expert labeling by medical professionals ensured

reliable and accurate ground truth data. Moreover, the LW-2DCNN architecture itself was thoughtfully designed and optimized explicitly to address the complexities associated with diagnosing four distinct pulmonary conditions: lung cancer, pneumothorax, pneumonia, and ILD. Collectively, these factors contribute significantly to the demonstrated robustness and diagnostic reliability of the proposed model.

#### D. HYBRID EXPLAINABLE FRAMEWORK ANALYSIS

The integration of ReciproCAM with the proposed LW\_2DCNN model generates saliency maps that provide valuable insights into the model's reasoning during prediction. A saliency map is a visualization technique that highlights regions of interest in an image from the model's perspective, aiding in understanding how the model identifies patterns and features for classification. Regions highlighted in red indicate areas with higher salience, meaning they contribute more to the model's prediction, while regions shaded in blue indicate lower salience.

In the case illustrated in Fig. 14, the ground truth is labeled as a lung with ILD disease. The LW-2DCNN model successfully predicted the class as ILD, after that ReciproCAM highlighted specific areas in red, corresponding to ILD partition in the patient's chest. That is the way our hybrid framework used to justify this ILD disease prediction, and similar for remaining classes as normal, lung cancer, pneumothorax, and pneumonia.

This visualization not only validates the model's prediction but also provides interpretability, enhancing trust in the model's decisions. By highlighting the key features contributing to each classification, the ReciproCAM technique can assist clinicians in understanding the rationale behind the model's predictions, making it a valuable tool for supporting decision-making in medical imaging.

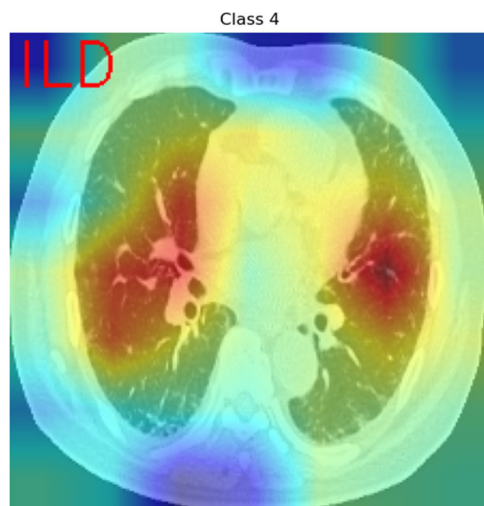


FIGURE 14. ReciproCAM explained for an ILD cases by saliency map.

#### E. COMPARISON TO PREVIOUS STATE OF THE ART

The results of our study have been systematically compared with previous literature focused on developing lung disease prediction models using chest CT imagery. Table 3 summarizes this comparative evaluation, highlighting performance metrics of our proposed LW-2DCNN model alongside several existing state-of-the-art methodologies. Utilizing standardized image dimensions of  $224 \times 224$ , our LW-2DCNN model achieved exceptional classification accuracy of 99.80% across five distinct pulmonary classes: normal, lung cancer, pneumothorax, pneumonia, and ILD.

Among prior approaches employing lightweight CNN-based architectures similar to ours, the Hybrid Metaheuristic Integrated CNN [46], COVID-FACT model [47], and the Lightweight DML model [49] reported Acc values of 98.91%, 90.82%, and 93.21%, respectively. Another related study [45] employing a fine-tuned ResNet50V2 model obtained Acc of 96.45%. Notably, the CYXNET methodology [48] closely approached our performance, reporting Acc of 99.77%. Nevertheless, our proposed LW-2DCNN consistently outperformed all referenced models in Acc. Furthermore, while existing studies typically focused on classifying between two to four pulmonary conditions, our model distinctly demonstrates the capacity to accurately differentiate among five classes, underlining its robustness and diagnostic versatility.

TABLE 3. Comparisons with previous state of the art.

| Research              | Method                                | Image size     | Number of classes | Acc (%)      |
|-----------------------|---------------------------------------|----------------|-------------------|--------------|
| [45]                  | ResNetV2                              | 224×224        | 4                 | 96.45        |
| [46]                  | Hybrid metaheuristic integrated CNN   | 512×512        | 3                 | 93.21        |
| [47]                  | COVID-FACT                            | 256×256        | 3                 | 90.82        |
| [48]                  | CTXNET                                | 32×32          | 3                 | 99.77        |
| [49]                  | Lightweight DLM                       | 488×408        | 2                 | 98.91        |
| <b>Proposed model</b> | <b>LW-2DCNN integrated ReciproCAM</b> | <b>224×224</b> | <b>5</b>          | <b>99.80</b> |

#### F. ABLATION STUDY

##### 1) CHANGING OPTIMIZER FOR THE PROPOSED MODEL

To comprehensively evaluate the impact of optimizer selection on our proposed LW-2DCNN architecture, we conducted an ablation study comparing the original Adam optimizer against two alternative optimizers: Stochastic Gradient Descent (SGD) and RMSProp. Fig. 15 illustrates the predictive performance when the Adam optimizer was substituted with SGD. While the original Adam-optimized model misclassified only 2 out of 1000 cases, exclusively within the lung cancer category, the model employing SGD misclassified 6 cases of lung cancer, 20 cases of pneumothorax, and 4 cases of pneumonia.

When utilizing RMSProp, as depicted in Fig. 16, the model performance improved relative to SGD, yet inaccuracies persisted with 4 lung cancer cases and 1 pneumothorax case misclassified. These results clearly indicate that the Adam optimizer is optimally suited to the proposed LW-2DCNN architecture, yielding superior diagnostic accuracy compared to SGD and RMSProp alternatives.

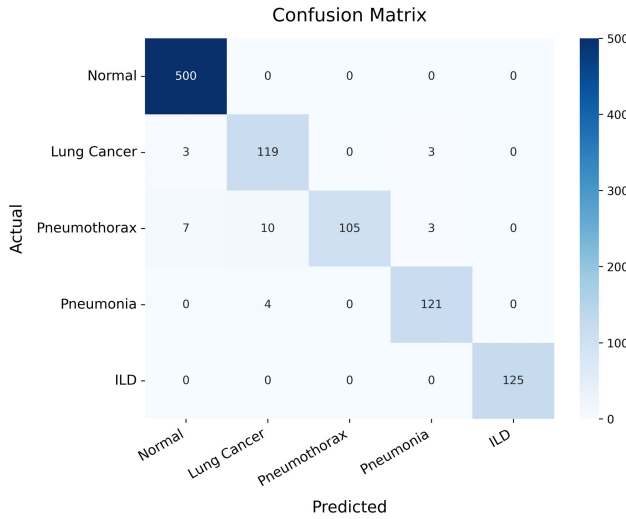


FIGURE 15. Performance of the LW-2DCNN with SGD optimizer.

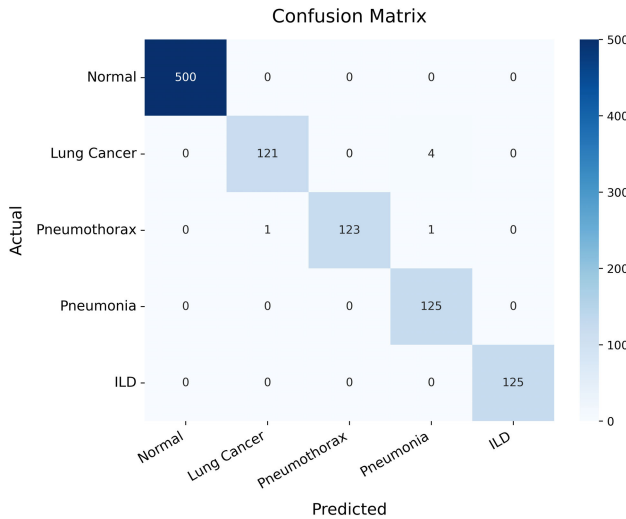


FIGURE 16. Performance of the LW-2DCNN with RMSProp optimizer.

## 2) COMPARING EXECUTION ON CPU-ONLY DEVICE

Another ablation study was conducted using an alternative computational setup comprising an AMD Ryzen 5 7600 CPU and 32GB DDR5 RAM, operating on Ubuntu 22.04 with TensorFlow 2.15. This secondary environment facilitated a comparative evaluation of performance differences between GPU-accelerated and CPU-only execution modes, specifically regarding runtime efficiency during classification and interpretability analyses using the LW-2DCNN model integrated with ReciproCAM. Additionally, we assessed the

comparative computational performance of the LW-2DCNN model integrated with ReciproCAM versus the same model employing GradCAM visualization under CPU-only conditions. Fig. 17 visualizes the execution time outcomes from these experiments.

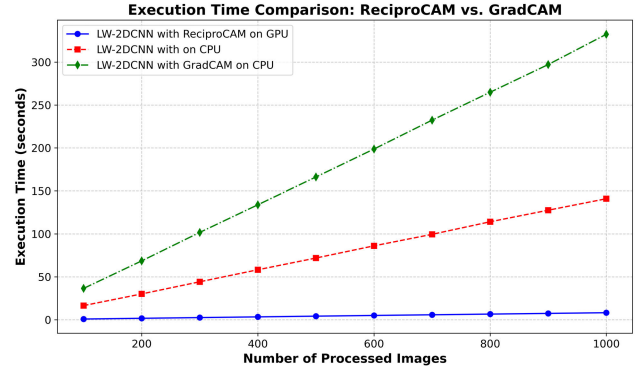


FIGURE 17. Testing time comparisons on CPU.

The GPU-enabled LW-2DCNN integrated with ReciproCAM demonstrated superior efficiency, completing predictions for 1000 images within seconds, whereas the same framework required approximately 150 seconds when executed exclusively on a CPU. Conversely, the LW-2DCNN framework utilizing GradCAM visualization exhibited significantly longer computational time, approximately 332 seconds, for processing the identical dataset of 1000 images. Extrapolating from these findings, it is evident from Fig. 17 that as dataset size increases, the computational advantage of LW-2DCNN combined with ReciproCAM over the GradCAM-integrated framework will become increasingly pronounced, highlighting the effectiveness and scalability of our proposed interpretability method.

## V. CONCLUSION

In this study, we introduced the LW-2DCNN, a lightweight two-dimensional CNN, integrated with ReciproCAM, achieving high accuracy and interpretability for classifying multiple lung diseases from chest CT imagery. The LW-2DCNN demonstrated superior diagnostic performance compared to state-of-the-art benchmarks, validated through various metrics including accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient. Optimizer ablation analyses established that Adam optimizer provided superior classification performance compared to alternatives such as SGD and RMSProp. Additionally, our performance analyses highlighted substantial computational efficiency advantages when using GPU acceleration over CPU-only execution.

A significant strength of our proposed framework is its computational efficiency. The LW-2DCNN integrated with ReciproCAM can predict and generate explainability visualizations for 1000 images within approximately 150 seconds on a CPU-only setup. This efficiency underscores the

practicality of our model in resource-constrained settings, enabling reliable lung disease diagnostics without requiring specialized GPU resources. Thus, our approach is particularly suitable for clinical environments where advanced computational resources may not be readily available.

However, our study is subject to certain limitations. First, the dataset employed was exclusively derived from a Korean population, which could potentially limit the model's generalizability to populations with diverse demographic and clinical characteristics. Additionally, the dataset did not include cases of COVID-19, which may restrict the robustness of diagnostic predictions in scenarios requiring differential diagnosis involving COVID-19 and related respiratory conditions.

Future research directions should aim to enhance model generalizability by validating performance on more diverse, balanced datasets, including representations of COVID-19 and other respiratory diseases. Furthermore, evaluations using multi-institutional and international datasets are recommended to validate the scalability and broad applicability of the LW-2DCNN and ReciproCAM framework. Extending this approach to other medical imaging modalities and exploring its integration within clinical workflows would further solidify the practical utility and clinical impact of this diagnostic tool.

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